

Hydraulic Aspects of Reservoir Operation and Flushing of Sediments for Steeply Sloping Perennial Himalayan Rivers

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Abstract:

The perennial rivers in hilly regions having steep slope like 1:10 to 1:50 pose very typical hydraulic problems. For such hydroelectric project, it is very difficult to create storages so as to cater for water requirement as well as creating space for deposition of large size sediments and boulders. Many challenges are faced while optimising the length and depth of reservoir. Whatever the space provided for deposition of such large sediments gets filled up in no time. Therefore, the spillway gates and intake of powerhouse has to be operated in tandem, so that the materials get washed away from the overflow spillway and simultaneously the required discharge is also drawn from the intake. During such operations, large sediment flow may enter through the power intake. Otherwise, the power intake needs to be closed for few hours to flush the reservoir intermittently. A case study was undertaken at Infraplan Hydraulic Laboratory, Pune, to study these aspects in mathematical model as well as physical model for one of the hydroelectric project in Uttarakhand. The attempt was made to optimise various hydraulic parameters and evolve efficient layout for spillway, considering the volume of water to be stored and flushing of the sediments. A 3D physical hydraulic model studies on a geometrical scale of 1:35 were conducted. Reservoir having a length of 0.6 km upstream was reproduced. Silt flushing tunnel was provided on the right bank for flushing of sediments just upstream of spillway. Numerical analysis and 2D mathematical model studies were conducted on Hec-RAS software. Various studies were conducted for optimizing the river slope to be dredged down and subsequent capacity of the reservoir. The studies were conducted for optimizing the flushing discharge and time for flushing and corresponding quantity of sediment flushed out from reservoir including large boulders. The present paper describes the optimization of the various hydraulic parameters such as reservoir capacity, amount of sediment flushed and time for flushing.

Keywords: Reservoir, sedimentation, reservoir drawdown flushing, flushing discharge, river dredging, Boulder movement.

1. Introduction

A significant part of India's hydropower potential exists in the Himalayan and North Eastern regions. The perennial rivers in these areas, with their large potential for hydraulic heads, offer promising opportunities for large-scale power generation. However, due to the fragile geology of the hills and the steep valley slopes, these rivers carry substantial sediment loads during the monsoon season. Sedimentation poses a serious challenge for dams located on Himalayan rivers, which experience heavy sediment flows. In hydropower project reservoirs, a major issue is the deposition of sediment and the need for effective, timely flushing, whether for storage or run-of-the-river projects. Over 80% of the annual sediment inflow occurs during the monsoon, making efficient sediment management during this period critical to extending the economic life of the reservoirs. This paper explores the general design and operational considerations of reservoir sedimentation and its flushing strategies. Also, a case study is presented based on the physical model studies carried out at Infraplan Hydraulic Laboratory, Pune.

2. Hydraulics of Steeply Sloping Himalayan Rivers

Rivers in the Himalayan region offer both challenges and opportunities for hydropower development due to their distinct hydraulic features, such as perennial flow, steep slopes and availability of high heads. These aspects also greatly influence flow velocities and sediment transport. All these factors should be considered during project design, as high velocities over the spillways can lead to cavitation damage and significant scour in the downstream river and its banks, regardless of the energy dissipation measures in place. The steeply sloping rivers carries large sediment load, particularly during flood events. Sediment is transported in the form of bedload (gravel, boulders, and sand), suspended load (fine silt and clay), and dissolved load (chemical compounds). The suspended sediment load as high as 10000 ppm during high floods is being experienced often. The movement of large boulders is also often observed during flood events. The high sediment load is a significant challenge for hydropower projects, as abrasive wear can damage turbines, intake structures, and other infrastructure. Additionally, sediment deposition in reservoirs can reduce the storage capacity over time, impacting the operational efficiency of dams. Photo 1 &

2 shows the presence of large boulders in the rivers at upstream reaches having steep slopes, whereas Photos 3 & 4 shows the sedimentation in the reservoir upstream of the dams. During the last decade, both mathematical and physical model studies have been conducted to evolve criteria for drawdown flushing. Many small and big projects have been provided with low level sluices for flushing of sediments accumulated upstream of dam. Naphta Jhakri is one of the projects where drawdown flushing has been employed over the years and thereby achieving the estimated power potential.



Photo 1&2. Rivers in upstream reaches carrying large boulders



Photo 3&4. Sediments deposited in the reservoir

3. Reservoir operation in steep sloping rivers

The operation of reservoirs with effective sediment management is essential for ensuring the long-term benefits of dams and reservoirs. Himalayan rivers, characterized by steep gradients and high rainfall, often experience rapid runoff, resulting in heavy sediment loads. Proper management is necessary to prevent reservoir siltation. Many of these rivers are also fed by glacial and snowmelt, leading to significant seasonal and yearly variations in both flow and sediment transport. Rivers with an average annual sediment concentration greater than 10 kg/m^3 and a sediment transport volume exceeding 100 million tons are typically classified as silt-laden rivers. Globally, 13 rivers transport more than 100 million tons of sediment annually, two of which are located in the Himalayan ranges of India, as illustrated in Table 1.

Table 1. Annual sediment discharge of Ganga and Brahmaputra rivers

River Name	Ganga River	Brahmaputra River
Watershed area ($10,000 \text{ km}^2$)	95.5	66.6
Annual runoff (million m^3)	371000	384000
Annual sediment discharge (million ton)	1451	726
Sediment concentration (kg/m^3)	3.92	1.89

The construction of reservoirs on steep-sloping rivers alters the sediment transport dynamics, impacting ultimately the reservoir operations. Optimizing reservoir management has become a key challenge in modern practice, aiming to sustain the sediment transport capacity of river channels. Numerical studies have examined the regulation of the water-sediment relationship in steep rivers. Adjusting reservoir operations based on real-time sediment monitoring can improve sediment management by regulating water release schedules or adjusting storage levels. Regular monitoring of sediment inflow and deposition rates enables timely adjustments to

operations and maintenance. Design strategies often include making reservoirs deeper and narrower to minimize sediment buildup, as sediments tend to settle in shallow, wide sections. Additionally, designing spillways to handle large volumes of sediment-laden water can help prevent sediment accumulation. Various reservoir operation policies are applied, such as the Standard Linear Operating Policy (SLOP), the concept of storage zoning, and rule curves.

4. Sediment Management techniques

Sedimentation in reservoirs leads to several negative impacts, not only within the reservoirs themselves but also in the upstream and downstream areas of dams. Some of the key issues caused by sedimentation and erosion processes include;

- Reduced storage capacity in reservoirs due to sediment accumulation.
- Increased flood levels upstream of the reservoir exceeding design estimates, caused by altered river slopes.
- In flood control dams and reservoirs, diminished storage capacity leads to changes in regulation and operational strategies, reducing the effectiveness of flow and flood management and increasing risk.
- Erosion of riverbanks and bed incision in downstream areas.
- Malfunction and clogging of hydromechanical equipment like flow control gates, sluice outlets, and vents.
- Abrasion and cavitation of concrete structures such as spillways, roller buckets, cut-off walls, sediment bypass tunnels, and channels.

A variety of sediment management techniques are available to maintain reservoir capacity and facilitate the passage of sediments downstream, many of which have been successfully implemented in some of the existing reservoirs. These sediment management methods can be categorized in several ways. One common approach is to group them based on the type of measure, such as structural, non-structural, and recurrent measures, as illustrated in Table 2.

Table 2: Sediment Management Techniques

Sediment Management Techniques		
Structural	Recurrent	Non-Structural
• Erosion control structures • Check dam • Sediment traps and retention basin • Sediment diversion tunnel and channel • Sediment diversion tunnel and channel • Guiding/ training structures (open levee, vanes, baffles, weirs, dikes, etc.) • Off channel reservoirs • Dam heightening	• Sluicing • Flushing • Density current venting • Sediment replenishment/ relocation • Hydraulic dredging/ syphoning/ slurry transport • Dry excavation/ trucking/ landfill	• Catchment/ land use management • Erosion control (afforestation, vegetation, contour farming) • Optimization of reservoir operation rule/ strategy • Sediment modelling • Real-time monitoring • Reservoir Morphology Information System

5. A Case Study

Among the numerous case studies on sediment management and reservoir operation, this paper focuses on a particular example of a steep sloping river located in the foothills of upper Himalayan ranges, which carries a huge sediment load. The project is proposed on one of the Perennial Himalayan River which has steep slope of the order of 1 in 10 in entire reach of reservoir to be studied. The project envisages construction of a concrete gravity dam. The river takes 90 degrees turn at chainage of around 450 m upstream of dam axis. The design flood is estimated to be 1850 m³/s. Two power intakes are proposed on the right bank of river to divert 25 m³/s discharge. One diversion tunnel (DT) cum silt flushing tunnel (SFT) is provided to flush the sediments in the reservoir with design discharge of 500 m³/s.

The numerical as well as physical model studies were carried out at Infraplan Hydraulic Laboratory, Pune, for optimizing the river slope and capacity of the reservoir. Numerical analysis and 2D mathematical model studies were conducted on Hec-RAS software which is widely used for hydraulic and hydrologic modeling. A 2D model was set up in Hec-RAS latest version of software. High resolution digital elevation terrain data prepared in GIS software was imported in RAS mapper. The river portion of 1,300 m upstream and 500 m downstream from dam was modelled as '2D flow area' in Hec-RAS. Resolution cell size of 10m x 10m was applied for River portion. The results of velocities, depths and water level were studied. The sedimentation profiles were also studied.

Also, a 3D physical comprehensive hydraulic model was constructed on a geometrical scale of 1:35 at IHL. Reservoir having a length of 600 m upstream covering the entire reach of the reservoir up to FRL was reproduced. Silt flushing tunnel was also reproduced on the right bank for flushing of sediments just upstream of spillway. The spillway, power intake and silt flushing tunnel models were fabricated in foam sheet and transparent perspex material. Various arrangements were provided for measurement of various parameters. Figure 1 shows the schematic diagram of model arrangements for physical studies.

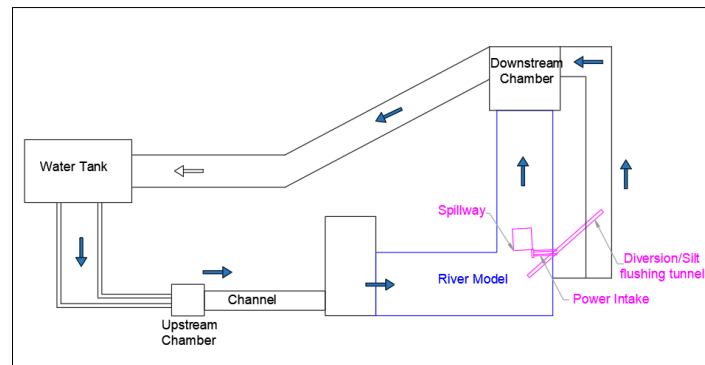


Figure 1. Model Arrangement Layout

Numerical model studies showed that the flow in the reservoir aligns to the right bank of river due to 90 degrees turn at upstream of dam axis. The velocities of the order of 8 m/s to 12 m/s were observed in the reservoir. The physical model studies were conducted for various reservoir water levels and corresponding discharges with various sediment concentration rates. The preliminary studies showed that the flow is taking a 90 degree turn and further aligning to the right bank. The flow conditions were supercritical in nature having high velocities of the order of 8 m/s to 12 m/s, just upstream of reservoir. The results from numerical model were similar to those of physical model. The comparison is shown in Photo 5 and Photo 6.

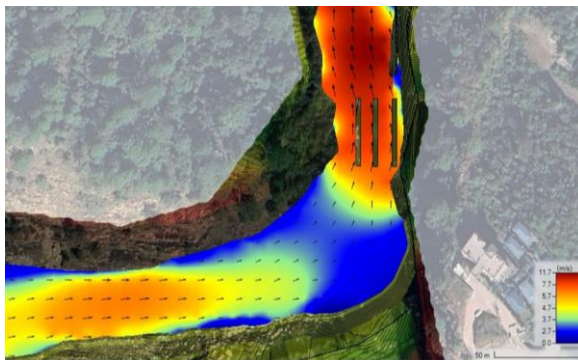


Photo 5. Flow conditions at upstream of spillway – Numerical studies



Photo 6. Flow conditions at upstream of spillway – Physical model studies

The studies were conducted for sediment management in the reservoir. Various alternatives were considered for sediment management. In that, drawdown flushing of reservoir technique was referred. The sediment deposition profiles were referred from the mathematical model for various years of deposition, such as 1, 2, 5 and 10 years. These sediment profiles were laid on the rigid bed model. The studies were conducted for flushing of these sediments for various flushing discharges and duration. Reservoir was surveyed after each experiment to assess the sediment flushing characteristics and the sediment quantities were assessed. The sediment simulation in the physical model was carried out based on the bed materials obtained from the sediment deposition pits on the site and presented in Table 3. The prototype sediment gradation shows sizes from 0.2 mm to more than 500 mm even upto 1 m size boulders, having higher percentage gradation of boulders. In order to simulate gradation curve,

different scaling criteria such as fall velocity as well as Shield's criteria were considered. Fall velocity criteria would be critical for reproducing the deposition of injected sediments on to the river bed, whereas the Shields criteria would be an important aspect for reproducing and determining the sediment movement in the model based on the tractive force. Different materials such as crushed sand, grit, along with materials having low specific gravity such as walnut shell powder, coconut shell powder was considered for simulation. In view of the higher sediment sizes the crushed sand along with grit was found to be preferable to reproduce the prototype gradation. The sediment sizes arrived for these scaling criteria are represented in Table 3. It can be observed that the sediment sizes required to reproduce below 1 mm of prototype sediment sizes were very fine and would not be feasible to reproduce using sand. The mixture of fine silt, sand and gravels of size having model particle diameter ranging from 0.2 mm to 15 mm were used to reproduce the sediments and they were mixed in the proportion so as to have a simulated gradation as per the prototype data.

Table 3. Sediment Gradation Curve

Sediment Sizes	% Passing	Sediment dia, mm, model as per Shields criteria	Sediment dia, mm, model as per Fall velocity criteria
0.0625	0.0%	0.0018	0.025
0.125	1.20%	0.0036	0.050
0.25	2.90%	0.0071	0.080
0.5	19.70%	0.014	0.114
1	19.70%	0.028	0.150
2	19.70%	0.057	0.196
4	36.70%	0.11	0.26
8	37.80%	0.22	0.37
16	42.40%	0.45	0.60
32	42.40%	0.91	1.00
64	44.50%	1.82	2.00
128	53.40%	3.65	3.67
256	73.70%	7.30	7.40
512	100.00%	14.60	14.50

The physical model was reproduced as a rigid bed model. The sedimented profiles arrived from mathematical model were laid over the rigid bed to reproduce the various years of deposition pattern. The drawdown sediment flushing studies were carried out by lowering the reservoir level for the various range of discharges and for different sediment concentration. During the flushing of the reservoir, sediments were also injected into the upstream end of model at a specified rate ranging from 5000 ppm to 10000 ppm. To assess the sediment removal through Silt flushing tunnel (SFT), initially the studies were conducted by lowering the reservoir level from FRL to MDDL and drawdown of reservoir was carried out by passing the flow through SFT. It was observed that very less amount of sediments were removed through SFT and its effect was very localized. To achieve the empty flushing, the reservoir level has to be lowered by opening both the radial gates. After opening the radial gates, the reservoir got emptied and the maximum flow as well as sediment movement was observed through the spillway. Photo 7 and 8 shows the flow conditions in the reservoir during flushing of sediment profile for a discharge of 100 m³/s and 200 m³/s. Figure 2 shows the contour map for sediment deposited after 24 hours of flushing with discharge of 200 m³/s.



Photo 7. Flow conditions during flushing for sediment profile for 100 cumec discharge



Photo 8. Flow conditions during flushing for 200 Cumec discharge

From the studies, it was observed that there was good conformity in the results obtained from numerical and physical model studies. It is evident from the table that the discharge of 200 m³/s for the duration of 24 hours was

found to be more effective in flushing the deposited sediment. However, very less sediment was passing through the silt flushing tunnel than anticipated due to its alignment with reference to the center of the river. It was also observed that the sediments that were removed were mostly below 200 mm or so. However, the sediment those were trapped with the larger sizes were also not completely washed away. The movement of boulders also pose a significant challenge, and it was opined that a rock trap may be provided at the upstream of the spillway.

Various alternatives were studied for optimizing the river slope and capacity of the reservoir. The studies provided a good insight for arriving the reservoir capacity and developing the sediment management strategy including the operational schedule. The following table summarizes the results of the studies carried out on a physical model.

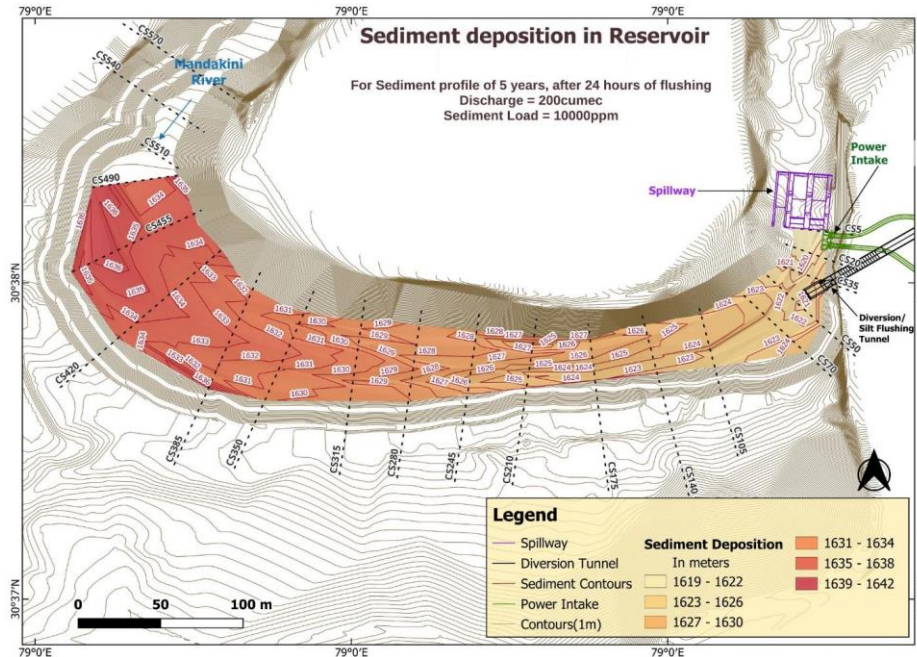


Figure 2. Sediment deposition for sediment profile of 5 year, $Q=200\text{m}^3/\text{s}$, Sediment load=10000 ppm

Table 4. Results of the studies

Discharge, Cumec	PPM	Year	Sediments Laid, m^3	Sediments Injected, m^3	Sediments Remained, m^3	Sediments Flushed, m^3	
						Through Gates	Through SFT
100	5,000	1	83,764	16,000	61,337	28,623	9,804
200	10,000	1	83,764	64,300	69,907	55,503	22,655
100	5,000	2	1,53,536	16,000	1,05,654	43,313	20,568
200	10,000	2	1,53,536	64,300	96,668	86,669	34,499
100	5,000	5	1,62,701	16,000	1,08,563	46,682	29,887
200	10,000	5	1,62,701	64,300	89,510	92,896	36,179
100	5,000	10	2,15,299	16,000	1,34,780	65,053	37,809
200	10,000	10	2,15,299	64,300	94,551	1,22,454	69,376

The results mentioned in above table indicates that substantial amount of sediment is washed away by drawdown flushing of the reservoir for an approximate period of 24 hours. The replenished capacity of the reservoir is more than 80%. The sediment got deposited at typical locations before and after the curvatures in the river portion amounts to the reduction in the reservoir capacity to the tune of 20%. This case study aptly illustrates the use of physical models, with sediment gradation curve simulated as per the prototype data.

The results of the mathematical model were in agreement with the physical model studies in respect of flow conditions, velocity distributions and pattern of depositions. The movement of silt and finer particles could be modelled in a mathematical model, however physical model was very useful to assess the movement of large size sediments and boulders.

The physical model showed that, due to upstream steep slope, the approach flow is pushing the formation of reservoir further downstream. Reservoir fluctuations of the order of 1 to 2 m were observed. The model was

further utilized to assess the effect of upstream river slope on the flow conditions as well as movement of boulders. So also, the dredging profile of the river was revised so as to have better flow conditions upstream of spillway. Further the model studies were carried out to evaluate the operation of spillway, intake and flushing gates for effective flushing of the sediments without hampering much on the reservoir capacity or with minimum shutdown of power plant.

6. Conclusion

Sedimentation in the steep sloping river poses many challenges in design and operation of the hydropower projects. Model studies provide a significant and dependable tool to assess the sedimentation and its management in the reservoir including flushing optimization. The integrated approach of coupling physical and mathematical models often proves complimentary. However, sediment studies being predominantly qualitative in nature and involving several parameters, the assessment of results require expertise, judgement and knowledge of actual site conditions.

7. Acknowledgement

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8. References

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